



Chihyeon Yun

AI Researcher & Engineer

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- ✦ **Multimodal Auto-Editing (M.S. & First Author):** Designed visual speaking-state and audio diarization late-fusion frameworks, utilizing Beam Search decoding to estimate optimal camera cuts (achieved 90.10% agreement with experts).
- ✦ **Spatial AI & Metaverse (ETRI R&D):** Engineered high-accuracy indoor Visual Positioning Systems (VPS) for cyber-physical space registration, and designed Visual-Language Model (VLM) zero-shot anomaly detection pipelines for real-time video semantics.
- ✦ **3D Reconstruction & VLA Robotics (PoC):** Coupled 2D MT-DUNet and Beer-Lambert radiative 3DGS for coronary artery 3D reconstruction under extreme memory optimization, and integrated SmoVLA control in Unity ROS simulation.

Research & Career Timeline

2025.02 - 2025.06

Electronics and Telecommunications Research Institute (ETRI)

Post-Master's Research Fellow at Hyper-Realistic Metaverse Lab. Worked on indoor visual positioning (VPS) coordinates mapping and VLM semantic anomaly detection. Managed joint government R&D milestones.

2023.03 - 2025.02

Kyungpook National University

Master of Science in Artificial Intelligence (GPA: 4.11/4.3). Conducted research in Multimodal Language & Cognition Lab on audio-visual late-fusion, speech diarization, and generative audio diffusion.

Core Tech Stack

Deep Learning & Audio

PyTorch Transformers DiffWave WhisperX
Pyannote.audio Librosa

Vision & Robotics

3DGS Visual SLAM/VPS MT-DUNet SmoVLA ROS/ROS2
Unity

Medical Imaging

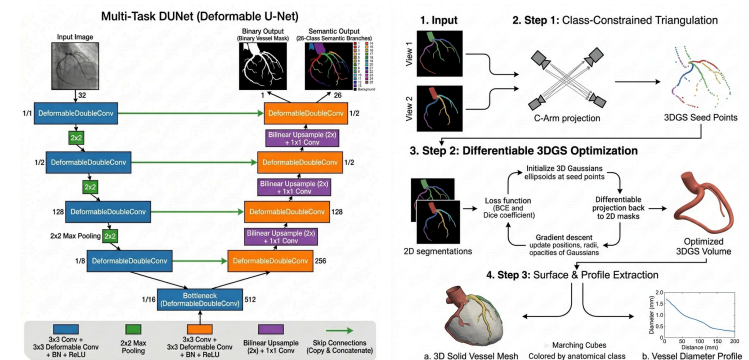
3D Reconstruction

LCA CAG 3DGS Reconstruction & Dashboard

A 3D volumetric reconstruction PoC that combines a 2D Multi-Task Deformable UNet (MT-DUNet) and Beer-Lambert radiative 3D Gaussian Splatting (3DGS) to recover 3D coronary artery structure, radius profile, and volume from sparse 2D C-arm X-ray angiograms.

- ♦ **MT-DUNet (DCNv2)**: Integrates deformable convolutions to adapt to complex vessel tortuosity, simultaneously predicting binary vessel masks and 25 anatomical branch labels.
- ♦ **Class-Constrained Triangulation**: Prevents ghost-point artifacts in epipolar matching by pairing identical anatomical branch labels across views, yielding a 4.5x denser centerline seed.
- ♦ **Beer-Lambert Radiative 3DGS**: Solves the transparency occlusion problem of X-ray projection by analytically integrating density along rays, optimizing vessel radius (r) and opacity (α).
- ♦ **High Efficiency**: Achieved 8x GPU memory saving (2.5 GiB execution) on an L4 GPU via downsampling and multi-resolution rendering.

Tech Stack: PyTorch, 3DGS, Deformable Conv (DCNv2), Python, HTML/JS



AI Agent

RAG Systems

Chihyeon.ai Personal RAG Assistant

A sophisticated personal RAG system that serves as an interactive AI secretary, providing high-fidelity, grounded answers based on academic research papers, CV, and project documentation.

- ◆ **High-Fidelity Parsing:** Leverages LlamaParse & Jina Reader API to flawlessly extract complex academic layouts, mathematical formulas, and JS-rendered web content.
- ◆ **Secure Serverless Proxy:** Uses Vercel Functions to encrypt API credentials and bypass region locks, ensuring safety and compliance.
- ◆ **Vector Storage:** Fast semantic matching over high-dim Pinecone space powered by Gemini 2.5 Flash for near-instant, streaming answers.

Tech Stack: Gemini 2.5 Flash, Pinecone (Serverless), Vercel Functions, LlamaParse, Jina Reader, Python



Robotics

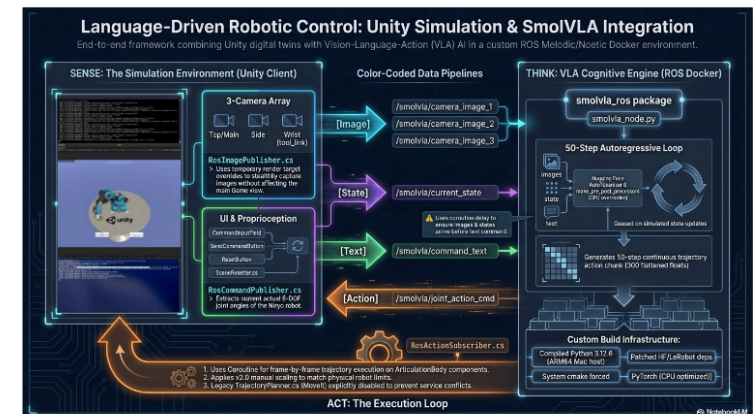
Simulation

Unity Pick & Place SmolVLA Integration

An end-to-end integration of the SmolVLA (Vision-Language-Action) model within a virtual physics environment to perform robotic 'Pick and Place' tasks.

- ✦ **Vibe Coding Paradigm:** Worked with LLMs to prototype network interfacing, accelerating boilerplate setup to focus on core robotics integration.
- ✦ **Communication Pipeline:** Built ROS-Unity bridges linking virtual cameras and joint torques for real-time physics simulation feedback.
- ✦ **Visual Control Loop:** Feeds camera images directly to SmolVLA agents for high-level spatial coordination and torque control verification.

Tech Stack: SmolVLA, Multimodal LLM, Unity Engine, ROS, Docker, PyTorch



01

Journal Submission (Pending / First Author)

Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis

Fuses active speaker detection (Swin Transformer) and diarization (WhisperX/Pyannote) visual-audio streams to align channels, decoding edits via a customized Beam Search logic matching 90.10% of expert visual pacing.

- Proposed a late-fusion multi-camera synchronization and editing framework, integrating speech state classification and visual cue matching into a Beam Search sequence decoder.

Tech Stack: Speaker State Estimation, Swin Transformer, WhisperX, Pyannote.audio, Beam Search, Python

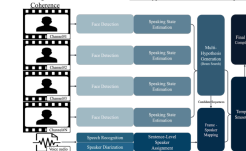
Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources

Introduction

- An automated system for editing multi-channel video by synchronizing raw inputs through separate visual and audio pipelines.
- Deep late-fusion of multimodal information for face recognition, speaking-state estimation, speech recognition and speaker identification.
- A multi-hypothesis editing approach that provides candidate versions based on probability distribution over time for extracting frames from each channel-stream.
- Frame-level match accuracies of 84.99% and 90.10% for the 3-channel and 5-channel videos, highlighting the system's robust performance across varying speaker complexities.

Method

- For multi-speaker, multi-channel content, we adopted an **intuitive approach** to make automated editing feasible. This is primarily because datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare.
- The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automating video editing.
- For the audio modality, our goal is to accurately detect active speech segments and identify the corresponding speakers from diverse and often noisy multi-channel sources - **Speech Recognition, Speaker Diarization**.
- The visual approach involves estimating the speaking state based on the facial region of the speaker - **Face Detection, Speaking State Estimation**.
- Integration and Channel Selection - **Multi-Hypothesis Generation for Temporal Alignment**.



Experiment & Result

- Dataset**
 - Audio recordings provided by a Korean public broadcasting company in South Korea. (3-channel video dataset, 5-channel video dataset)
- Evaluation**
 - Compare the ground-truth labeled data with output generated by our system at the frame level.
 - For the Quantitative Analysis, Analyze the frame-level alignment accuracy under various smoothing and temporal conditions.

Table 1: Frame-level alignment accuracy (%)

Condition	3-Channel	5-Channel
Raw Data	84.99	90.10
Smoothed Data	85.12	90.25
Temporal Shift	84.85	89.95

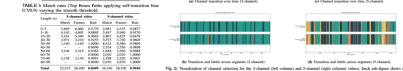
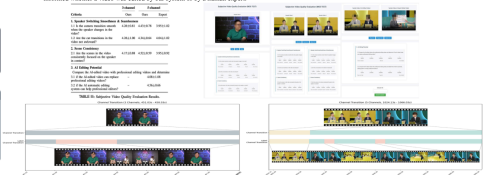


Fig. 2. Visualizations of channel selection for the 3-channel (left column) and 5-channel (right column) videos. Back-left figure shows the channel selection and probability for the 3-channel video.

- For the Qualitative Analysis, Analyze subjective assessments of editing quality and viewer experience.
- The analysis was performed with 24 participants, including developers and broadcasting industry professionals, under a blind setup, where they were not informed whether a video was edited by our system or by a human expert.



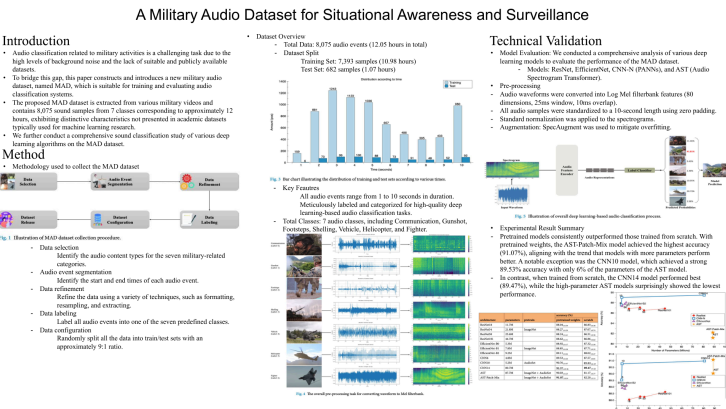
02 Scientific Data (Nature Portfolio, SCIE, 2024 / Co-Author)

A Military Audio Dataset for Situational Awareness and Surveillance

Curated and open-sourced 8,075 military audio clips (MAD) representing 7 hazard classes for security surveillance, designing raw audio preprocessing filters and evaluating AST models achieving 91.07% classification accuracy.

- ✦ Built audio feature extraction pipelines (Librosa log-Mel filterbanks) and structured AST-Patch-Mix classifier benchmarks for real-time audio defense surveillance.

Tech Stack: Audio Spectrogram Transformer (AST), Mel Filterbank, Python, Librosa



03

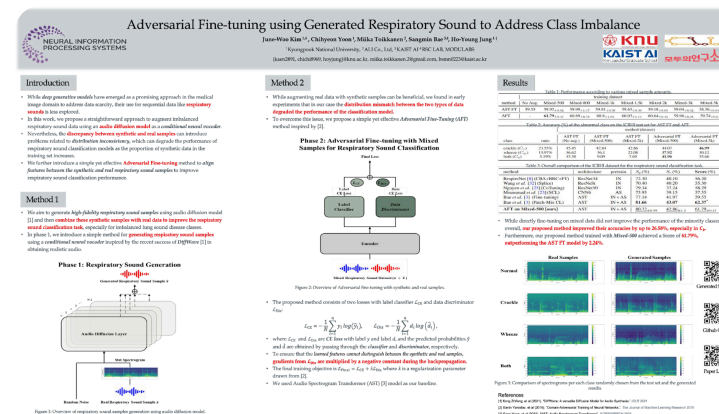
NeurIPS 2023 DGM4H Workshop (International Conference / Co-Author)

Adversarial Fine-tuning using Generated Respiratory Sound

Tackled medical data scarcity via DiffWave diffusion synthesis, training deep classification structures with domain-adversarial fine-tuning (AFT) loops to bridge synthetic and real sample sets, increasing minority class recall by 26.58%.

- Designed medical sound synthesis pipelines using conditional audio diffusion neural vocoders, and proposed domain-adversarial fine-tuning for diagnostic label alignment.

Tech Stack: DiffWave, Domain Adversarial Training (DAT), Audio Synthesis, PyTorch



04

KCC Proceedings (Domestic Conference / Bronze Prize)

CutMix-SupCon: Contrastive Learning for Multimodal Emotion Recognition

Designed latent-level CutMix feature interpolation to merge KLUE-RoBERTa (text) and wav2vec 2.0 (audio) embeddings, optimizing Supervised Contrastive Learning (SupCon) for cosine similarity alignment on minor classes.

- Implemented multimodal late-fusion and contrastive learning alignments, improving speaker emotion recognition rates on class-imbalanced benchmark sets.

Tech Stack: wav2vec 2.0, KLUE-RoBERTa, Supervised Contrastive (SupCon), CutMix

CutMix-SupCon: Supervised Contrastive Learning with CutMix Augmentation for Multimodal-based Emotion Recognition

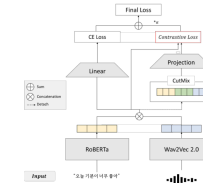
서론

- 멀티모달 데이터는 구조적으로 특징하기 어려운 문제가 있으며, 무작위 데이터 샘플의 문제로 인해 학습된 임베딩 기반 네트워크의 성능이 저하되어서는 안되며, 현재까지 존재하지 않는다.
- 현재 연구는 실제 환경의 무작위 데이터 문제는 고려하지 않고, 멀티모달 관련 문제에 대한 연구는 부족한 실정이다.
- 본 연구에서는 멀티모달 데이터의 불균형 문제를 해결하기 위해 CutMix-SupCon 방법을 제안한다. 제안된 CutMix-SupCon 방법은 실제 환경에서 추출된 음성-텍스트 모달 간의 결합된 특징표현으로부터 혼합된 특징표를 만들고, 혼합 비율을 통해 음성-텍스트의 결합된 멀티모달 특징 표현이 원본과 얼마나 유사한지에 대한 특징 함수를 통해 최적화하였다.

방법

- 제안하는 멀티모달 CutMix-SupCon 방법은
- 멀티모달 특징표현 결합
- CutMix 기반 멀티모달 혼합 방법
- SupCon

으로 나타내며, 전체 흐름도는 다음과 같다.



- CutMix 기반 멀티모달 혼합 방법
멀티모달 특징표현 x 와 y 에 대하여 샘플 $(x_{\alpha}, y_{\alpha}), (x_{\beta}, y_{\beta})$ 를 구성할 수 있고, 이후의 새로운 샘플 (x, y) 를 아래 식에 따라 만들 수 있다.
$$x = \alpha x_{\alpha} + (1 - \alpha) x_{\beta}$$
$$y = \alpha y_{\alpha} + (1 - \alpha) y_{\beta}$$
- CutMix-SupCon
CutMix 기반으로부터 결합된 멀티모달 특징표현을 위한 적절한 손실 함수를 적용하기 위해, 실제 공간에서 혼합된 정보를 구분하여 효과적인 학습을 가능하게 하는 CutMix-SupCon 방법을 제안한다.
$$L_{CL} = \sum_{i \in P} \sum_{j \in N} \log \frac{\exp(\langle x_i, x_j \rangle)}{\sum_{k \in N} \exp(\langle x_i, x_k \rangle)}$$
$$L = L_{CL} + L_{CE}$$

위와 같은 최종 손실 함수를 통해 모델은 혼합된 멀티모달 특징표현이 원본의 샘플과 고충 정보를 얼마나 포함하는지를 학습한다.

실험 및 결과

- 데이터셋: KEMDy19 Tense-Audio 멀티모달 감정인지 데이터셋
중국어 개입, 음성인식, 개입, 2018, 이상의 음성 개입 제거 등의 전처리를 통해 10,214개의 데이터 샘플을 확보
- 방법: 각 모달에 대한 특징표현 기법의 CE, Mixup, CutMix, 제안하는 CutMix-SupCon 방법에 대한 결과 비교

Input	Method	Acc	F1-score
Text		67.6	67.6
Text	Cross-Entropy	73.63	69.33
Audio		72.28	61.23
Text+Audio		73.37	69.09
Text	Mixup	71.28	67.27
Audio		71.48	65.59
Text+Audio		72.39	68.93
Text	CutMix	72.93	69.37
Audio		71.79	65.83
Text+Audio		74.74	70.77
Text	CutMix-SupCon	72.49	69.75
Audio		72.54	68.27
Text+Audio		75.33	73.17

표 1. KEMDy19 데이터셋을 사용한 감정인지 평가 결과

Input	Method	Weighted F1-score
	Cross-Entropy	73.00
Text+Audio	Mixup	72.98
	CutMix	74.77
	CutMix-SupCon	74.96



그림 2. KEMDy19 테스트 데이터 세팅에 대한 CE와 CutMix-SupCon의 예측 확률 분포 비교